White Paper

A New, AI-Augmented Horizon for Safety and Pharmacovigilance: Expert Predictions for 2026

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Introduction

The question of *if* AI can improve safety and regulatory workflows has been answered with a resounding *yes*. Now, organizations must determine how to integrate this technology to maximize efficiency, ease the burden on their teams, and ensure regulatory compliance. In a recent panel discussion, five IQVIA experts sat down to share their predictions for how safety and regulatory workflows will evolve in 2026, providing their expert guidance on the importance of keeping a human-in-the-loop, the role of AI governance in maintaining compliance, and how best to train staff for this major technology shift. New trends are on the horizon: leverage these expert insights to help your organization ride the AI wave to strategic success.



Looking back at 2025, what was the most surprising development in your area of expertise?

Archana Hegde, Senior Director, PV Systems & Innovations, Lifecycle Safety: The most surprising aspect is the cultural change we are seeing amongst the process owners, CROs, and end users of technology for pharmacovigilance (PV). We've seen a shift in terms of their participation, from not being able to digest all the technology to providing input on developing technology. The second is regulatory responses from the FDA, EMA, MHRA (the UK's Medicines and Healthcare products Regulatory Agency), etc. They've given us guidance on how to use AI and what our frameworks should look like.

Ian Fisher, Head of Development Analytics, Regulatory Affairs and Drug Development Solutions: We've seen an increase in the transparency around how regulatory decisions are made. One big surprise is the FDA's release of the Complete Response Letters (CRLs) and the clarity in terms of: these are the hurdles people have had to face to make it to market. If using AI tools, especially agentic AI, and asking questions from different perspectives, we can draw out valuable nuggets of information from CRLs, summary bases of approvals, and EPARs (European Public Assessment Reports) to increase success rates.

Michelle Gyzen, Senior Director, Strategic Regulatory Solutions; Head of Regulatory Services Innovation & Technology: It was the speed of AI normalization in regulatory. As many of you know, regulatory does not jump on the bandwagon quickly, but we moved from cautious pilots to full-blown expectations. Regulators are now assuming that AI is part of the operating model. What surprised me most is how global that shift was. It's not just the FDA and EMA; it's APAC (Asia Pacific). Health authorities are asking not just if you're using AI but how you're governing it. This has forced a lot of organizations and regulatory teams to evolve their tech fluency in 2025.

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— Michelle Gyzen

Louise Molloy, Director Medical Information & Pharmacovigilance: With advancements in technology, there's been a repositioning of the value of medical information. It used to be perceived as an add-on service. Now, it's being recognized as a source of strategic insights. In gathering medical information, we have direct contact with patients, HCPs, and caregivers that provide real-world insight. Over the past year, people have been exploring how to use that insight to inform labeling and ensure it is fit for purpose. The voice of the consumer is being brought into the conversation.

Ana Pedro Jesuíno, Director of Marketed Product Safety: It was the rapid integration of AI-driven tools within PV scope and oversight to increase efficiency and organize reliable, high-quality data. Yesterday, we were talking about how to integrate AI, and today, it seems that we never work without AI integrated. Regulators are promoting and driving the integration of AI within PV activities.



What is the biggest misconception around AI's role in compliance?

Gyzen: The biggest misconception in regulatory is that AI is magical or mysterious. It is neither. The real value of AI, especially in compliance and PV, is augmentation. We are surfacing insights faster and spotting challenges earlier, but we still need governance, transparency, and definition around the context of use. If you cannot explain how your AI came to a decision, it does not belong in a compliance system.

Molloy: With medical information, technology is augmenting the value of our expertise. We need more than just a human-in-the-loop; we need humans throughout the whole process to enable quality by design (QbD). It's important to acknowledge the anxiety around AI, especially with patients and customers, and emphasize that this happens with every new technology. It may be new technology, but we can learn this new language to ensure accountability.

Fisher: The biggest misconception has been that AI will make things easier. If used correctly, we should be looking at AI tools to help us reach better and more well-rounded decisions, especially if we prompt GenAI tools to consider different stakeholder perspectives. Instead of seeking information to bolster your existing belief, you can look for challenges before they arrive. Make sure you're battle hardened by understanding the potential pitfalls of your plans. AI is helping us unlock insights that were previously at the end of a long rabbit hole of PDFs.

Hegde: AI is not an autonomous decision-maker; it's an assistant. We cannot blame AI for inaccuracies. Accountability is still with the qualified professional. It doesn't eliminate risk; in fact, it brings in other areas [of risk] in terms of data integrity, AI outputs, or human management. It's not a no-touch approach; it's a smart-touch approach.



"How do you monitor the AI outputs and use feedback to improve the application? Invest in the people who will be using it and bring the right expertise to the table."

— Archana Hegde

What are the critical capabilities regulatory intelligence tools should integrate to support more harmonized compliance?

Gyzen: From a global perspective, there are three things to prioritize. First and foremost, multilingual comprehension tools that can parse regulations across formats and languages. Secondly, dynamic crosswalks that map new guidance and global frameworks to internal SOPs in real time and enable cross-regional capability. The third is explainability. We need audit-ready outputs with traceable logic — bonus points if they integrate into team workflows. If your regulatory intel lives on a dashboard that nobody opens, it's not intelligence, it's shelfware.

The FDA's draft guidance on AI in drug development has introduced concepts like risk-based validation and context of use. What practical steps should PV teams take to operationalize these expectations in 2026?

Hegde: The FDA guidance notes that validation should be based on the risks involved in a process. For any new technology being implemented, you should know exactly what it is used for and what functions it touches. You should define the context of use upfront, rather than figuring it out as you go. There also needs to be a strong governance structure to approve what goes into production, how it goes into production, and the checks and balances that need to happen before and after it goes live. From there, you need to determine your approach to lifecycle monitoring.

How do you monitor the AI outputs and use feedback to improve the application? Invest in the people who will be using it and bring the right expertise to the table. Bringing these solutions to fruition requires a combination of technology and process experts.





How should a team help prevent AI from making mistakes in safety reporting?

Hegde: We cannot stress enough how much humans need to be in the loop. Assume AI output is just a draft and never presume it is 100% accurate. The SME who has been doing the reporting must review the reports for correctness and AI bias as well as conduct QC and validation. From there, it can be submitted in, hopefully, a much shorter time span.

Jesuíno: We work in a highly regulated environment and must apply risk management to everything we do. If we are adding a new tool to our GxP system, we need to ensure it works well, validate it, closely monitor it, and keep humans in the loop. This is the way we should treat AI implementation in existing systems.

Molloy: As healthcare professionals, we know the responsibility that we have. To instill confidence in any new process, you must train people. We must focus on upskilling our experts and teaching them this new skill set. Empower team members to develop critical thought in validation so that they feel confident and supported.

What confidentiality considerations are important when AI is using and analyzing patient data?

Molloy: Our approach to GDPR (the EU's General Data Protection Regulation), confidentiality, and sensitive information still applies with AI. Medical information is a source of unstructured data. We wouldn't dream of setting up call centers without confidentiality and privacy disclosures, so why not approach AI the same way? Even if you haven't fully implemented AI, a disclosure statement is an elegant solution that helps the customer understand how you treat their data and that your company is an entity they can trust.

As AI and large language models become standard in PV and compliance, what are some practical steps organizations can take to demonstrate model validation, monitoring, and risk assessment to regulators?

Hegde: The FDA's Emerging Drug Safety Technology Program (EDSTP) allows organizations to bring their AI approach to regulators before implementation to get feedback. In addition to regulators, the forum includes other pharma companies and CROs and meets on a regular basis.

Gyzen: Regulators don't need AI to be perfect; they need it to be accountable. In terms of validation, we ensure that we make the logic visible, which has to be the new standard.



Are there any new oversight skillsets that Local Qualified Persons for Pharmacovigilance (LQPPV) will need in 2026 as AI continues to become more embedded in safety workflows?

Jesuíno: If LQPPVs are to ensure compliance and oversight, they need to understand how an AI algorithm works so that they can anticipate risks and corrective and preventative measures. They'll need to become experts in change control and understand what could trigger revalidation. It is fundamental to establish an effective governance model that integrates the different stakeholders in the system and keeps everyone on the same page.

In documentation, PV System Master Files (PSMFs) will start to describe the AI tools and their validation models. If companies are working with vendors, I would expect to see vendor certifications of any tools they use and an explanation of how changes are communicated to ensure effective oversight.

What are the most important ways that teams should involve their Target Product Profiles (TPPs) to drive alignment, inspection readiness, and long-term value across development and commercialization?

Fisher: TPPs define what a drug developer wants to achieve with their drug at launch, including intended patient outcomes and the magnitude of benefit compared to existing standards of care. There's still a requirement for an evolution towards multiple scenario-based TPPs. We're moving to a place where we can start to consider event-based scenarios from the earliest stages of development to make more informed inferences around which risk-benefit profile will be acceptable to regulatory agencies and desirable for patients and even payers.

We also need to start establishing a better understanding of what the unmet need is and what moves someone from an established standard of care to a new drug. This entails evaluating epidemiology, patient-reported outcomes, real patient burden, and access to generate your commercial TPP.

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— Ian Fisher



What transparency standards and safeguards should be in place to ensure ethical use of AI in patient-facing interactions?

Molloy: We need to focus on ethics and governance. In medical information, most data is spontaneous and unstructured. That implies a huge responsibility for our confidentiality and how we handle that data, which affects all the other downstream processes, including regulatory and PV.

In addition to disclosures, we need to establish guardrails around the adoption of technology. While we want processes to be lean and efficient, we also need to thoroughly understand them before we drop in new technology. If you implement an AI-based QC step to increase efficiency x10, what if AI makes a mistake, causing error propagation x10? While it's tempting to go as fast as possible, you need to employ an ethical mindset that focuses on understanding the process, building trust, and having a governance framework in place before you implement AI.

How can smaller pharma companies keep up with all these changes?

Gyzen: Smaller companies are in an outstanding position. Agility is an advantage. They can adopt modular AI platforms faster, setting up tools with built-in global frameworks, alerts, and monitoring. But I also caution that AI is not always needed. You don't need a hammer to do intricate work. In some situations, point solution automations are your friend. For small organizations, I recommend outsourcing selectively. Work with partners with major infrastructure and automate where it counts. Be adaptable and build smart systems.

"In medical information, most data is spontaneous and unstructured. That implies a huge responsibility for our confidentiality and how we handle that data."

— Louise Molloy

What safety or regulatory trend will define 2026? How should pharma teams prepare for it?

Fisher: Success in regulatory interactions will come down to being cognizant of what is expected. There is an incredible amount of transparency around expectations. It's very clear that the inefficiencies and delays in drug development arise when people push the boundaries of what is allowed with the intent of seeking acceleration. In 2026, go with the flow, be lean, strategic, directed, and clear on your trial design. Use whatever AI tools you have available to investigate how your approach might be received given regulatory precedence.

Jesuíno: The next step is that AI will become part of the scope of regulatory inspections. To prepare, develop a step-by-step implementation strategy that utilizes experienced vendors and/or validated tools on the market.

Hegde: We will see an emergence of hybrid roles; people with process knowledge that understand technology and vice versa will be successful. New roles will emerge, including people who define new PV processes and determine what types of prompts need to be written for AI in different functions. New talent is entering with new degrees, but we'll also need to upskill existing talent to prepare them to play senior roles, determine what processes will look like, and support compliance.

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— Ana Pedro Jesuíno



About IQVIA

IQVIA (NYSE:IQV) is a leading global provider of clinical research services, commercial insights and healthcare intelligence to the life sciences and healthcare industries. IQVIA's portfolio of solutions are powered by IQVIA Connected Intelligence™ to deliver actionable insights and accelerate innovations. With approximately 91,000 employees in over 100 countries, IQVIA is dedicated to accelerating the development and commercialization of innovative medical treatments to help improve patient outcomes and population health worldwide.
Learn more at www.iqvia.com.

About the authors



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Ian Fisher leads a team of experienced analysts to provide drug developers with insights and direction for indication prioritization, target product profiles, clinical development planning, and asset valuation. Ian and his team leverage structured and unstructured data and information within structured decision-making frameworks to increase the efficiency, transparency and success of strategic program development. He has over 22 years of experience in drug development and holds two master's degrees: in Chemistry and Health Economics.



MICHELLE GYZEN
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Michelle Gyzen has over 20 years of experience in the pharmaceutical, biotech, and medical device industries. She is a Senior Director with expertise in designing large-scale regulatory outsourcing programs, offshore resource modeling, and regulatory tech integration and automation. In her current role, Michelle is involved in designing strategic solutions for regulatory compliance, with a focus on operational efficiency, scalability, and technology.



ARCHANA HEGDE
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Archana Hegde joined IQVIA in 2015 and has held various leadership roles implementing technology solutions to support critical business processes. In her current role, she provides senior leadership and oversight for the Safety Tech-Service Integrated Offering delivery, combining IQVIA's expertise from a technology and service perspective while focusing on accuracy, efficiency, compliance, standardization, and automation.



ANA PEDRO JESUÍNO
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Ana Pedro Jesuíno is a seasoned leader in global pharmacovigilance and regulatory compliance, and is currently Global Head for IQVIA's Local QPPV Network. With over 15 years' experience in CRO and pharmaceutical industries, Ana has overseen safety systems in over 50 markets, driving process harmonization and inspection readiness. She holds a master's degree in Pharmaceutical Sciences and leads cross-functional teams and global training programs.



LOUISE MOLLOY
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Louise Molloy manages operations for the Medical Information & Pharmacovigilance team at IQVIA, focusing on optimizing processes and leveraging technology to enhance efficiency and compliance. Louise is an integral part of the global team, advising pharmaceutical companies on best practices, innovation, and automation in MI. She frequently speaks on topics such as integrating AI in medical information systems and the evolving landscape of pharmacovigilance. Located in Spain, Louise brings over 15 years of experience in the pharmaceutical industry.

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